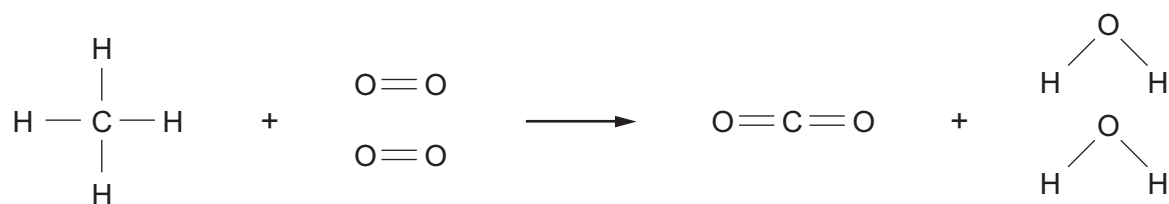


2. The burning of methane in air can be represented by the following equation.



The bond energies are given in the table below.

Bond	Bond energy (kJ)
C — H	413
O = O	498
O — H	464
C = O	805

- (a) Use the bond energy values to calculate the energy released when **all** the bonds in the carbon dioxide and water molecules are formed. [2]

Energy released = kJ



Examiner
only

(b) The energy needed to break **all** the bonds in the methane and oxygen molecules is 2648 kJ.

Calculate the overall energy change for this reaction and use this value to explain why the reaction is exothermic. [2]

Overall energy change = kJ

.....

.....

4

3430UE01
07



2. (a) Magnesium reacts with oxygen to form magnesium oxide.

Using the electronic structures below, draw dot and cross diagrams to show how bonding takes place during the formation of magnesium oxide. Include the electronic structures of the ions formed. [3]

magnesium = 2,8,2 oxygen = 2,6

(b) The melting points of magnesium oxide and sodium chloride are given below.

Substance	Melting point (°C)
sodium chloride	801
magnesium oxide	2852

Explain why there is a difference in their melting points even though they are both ionic substances. [2]

.....

.....

.....

.....



Examiner
only

(c) Using the electronic structures below, draw a dot and cross diagram to show the bonding in a molecule of carbon dioxide, CO₂. [2]

carbon = 2,4 oxygen = 2,6

7

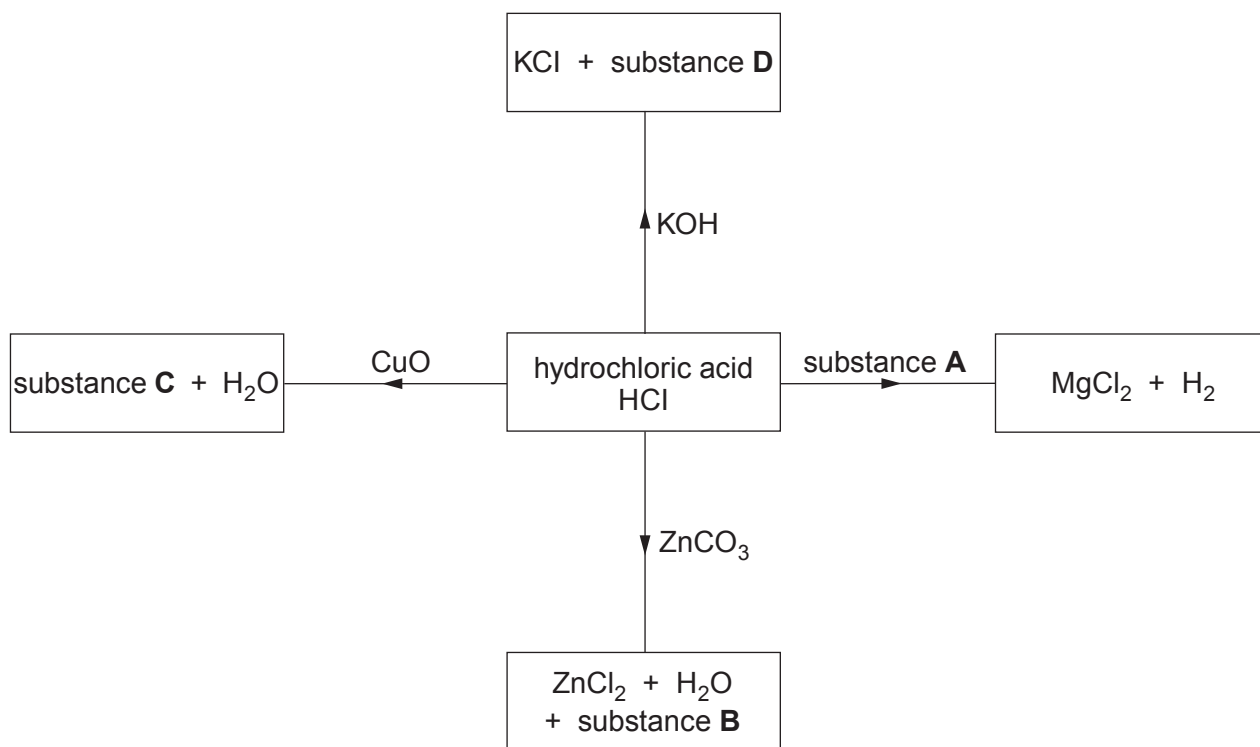
3430UE01
09



2. The reactions of acids with metals, bases and carbonates are summarised in the following equations.

- acid + metal $\rightarrow$ salt + hydrogen
- acid + base $\rightarrow$ salt + water
- acid + carbonate $\rightarrow$ salt + water + carbon dioxide

(a) The diagram shows some reactions of hydrochloric acid, HCl.



(i) Give the **names** of substances **A** and **B**.

[2]

Substance **A**

Substance **B**

(ii) Give the **formulae** of substances **C** and **D**.

[2]

Substance **C**

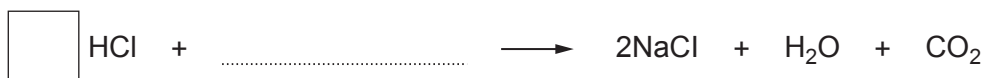
Substance **D**



- (b) Complete the equation for the reaction between hydrochloric acid and sodium carbonate by

- writing the formula of sodium carbonate on the dotted line
- putting a number into the box to balance the equation

[2]



- (c) Silver nitrate solution is used to identify the chloride ions present in hydrochloric acid.

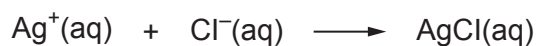
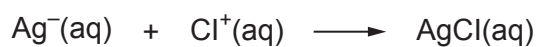
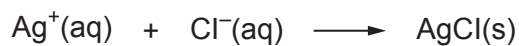
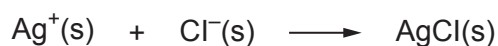
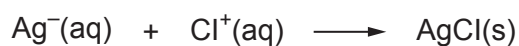
- (i) Give the observation made when silver nitrate solution is added to hydrochloric acid.

[1]

.....

- (ii) Put a tick (✓) in the box next to the correct ionic equation for the reaction between silver nitrate and hydrochloric acid.

[1]

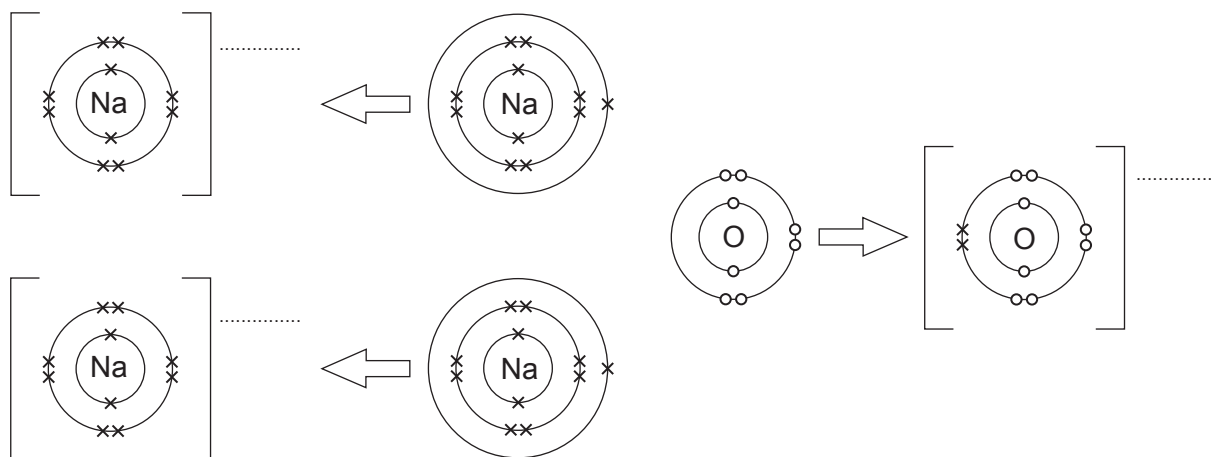

☐

☐

☐

☐

☐


2. (a) The table gives the electronic structures of sodium and oxygen.

Element	Electronic structure
sodium	2,8,1
oxygen	2,6

- (i) Sodium reacts with oxygen to give sodium oxide.

The diagram below can be used to show the electronic changes that occur as sodium oxide is formed.



Complete the diagram by

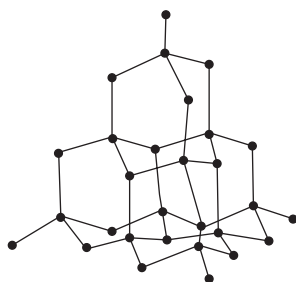
- **drawing arrows** to show the movement of electrons between the sodium and oxygen atoms,
- giving the **charges** of the sodium and oxide ions that are formed.

[2]

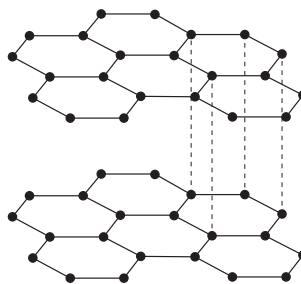


- (ii) Draw a dot and cross diagram to show the bonding in an oxygen molecule, O_2 . [2]

- (b) (i) Diamond and graphite are both made from carbon atoms. Their structures are shown below.



diamond



graphite

- I. State which structure is able to conduct electricity. Give a reason for your answer. [1]

.....

.....

- II. State which structure is used as a lubricant. Give a reason for your answer. [1]

.....

.....

- (ii) Give the name of another form of carbon that is used in drug delivery systems. [1]

.....

